

Shell PLC: ESG Performance Analysis 2016–2025

Data Appendix — Supporting Visualisations

Independent analysis | April 2025 | Data sourced from Shell PLC public sustainability disclosures

This appendix contains all 15 year-on-year visualisations generated from Shell's public sustainability disclosures (2016–2025). Charts are organised by ESG pillar and sub-theme. All analysis was conducted independently in RStudio; data was sourced from Shell annual sustainability reports and public ESG filings. No external assurance was applied.

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- G1 — Governance Indicators (Low-Carbon Capex; CEO Pay Ratio)

E1 — Scope 1 Emissions

Direct Scope 1 vs Equity Boundary Scope 1

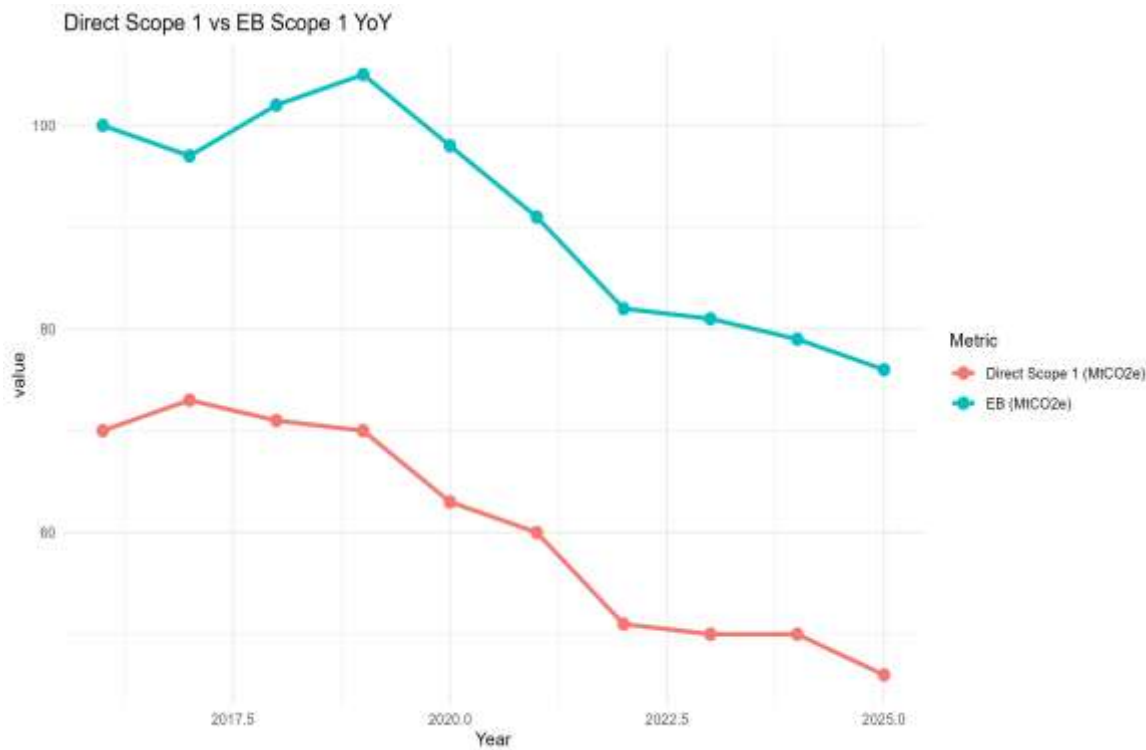


Figure 1. Direct Scope 1 vs Equity Boundary Scope 1 (MtCO₂ e), 2016–2025

Note: The persistent gap between Direct and Equity Boundary Scope 1 is consistent with asset divestment contributing to reported reductions alongside operational improvements.

Scope 1 Gas Breakdown: CO₂ , CH₄ , HCFs

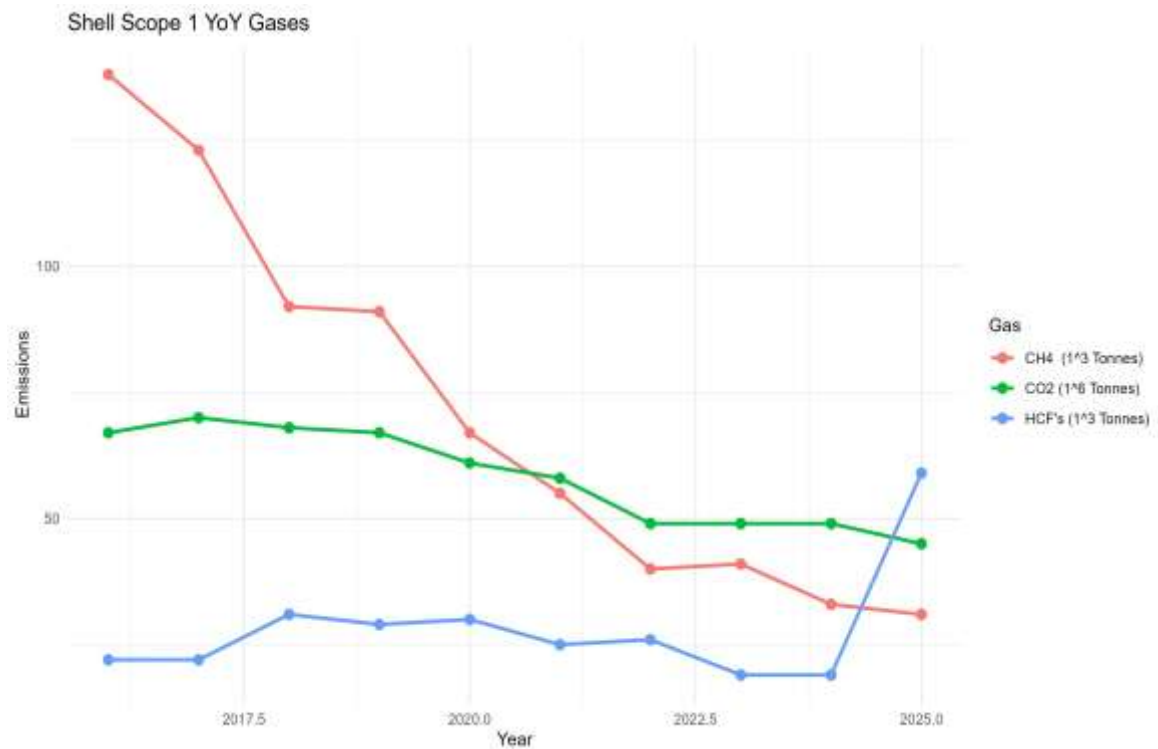


Figure 2. Scope 1 Gas Breakdown: CO₂ , CH₄ , HCFs (2016–2025)

Note: CH₄ (methane) recorded the sharpest decline, reaching 0.04% intensity — well below the 0.2% industry threshold. The 2025 HCF spike warrants monitoring.

Net Carbon Intensity (NCI) of Sold Products

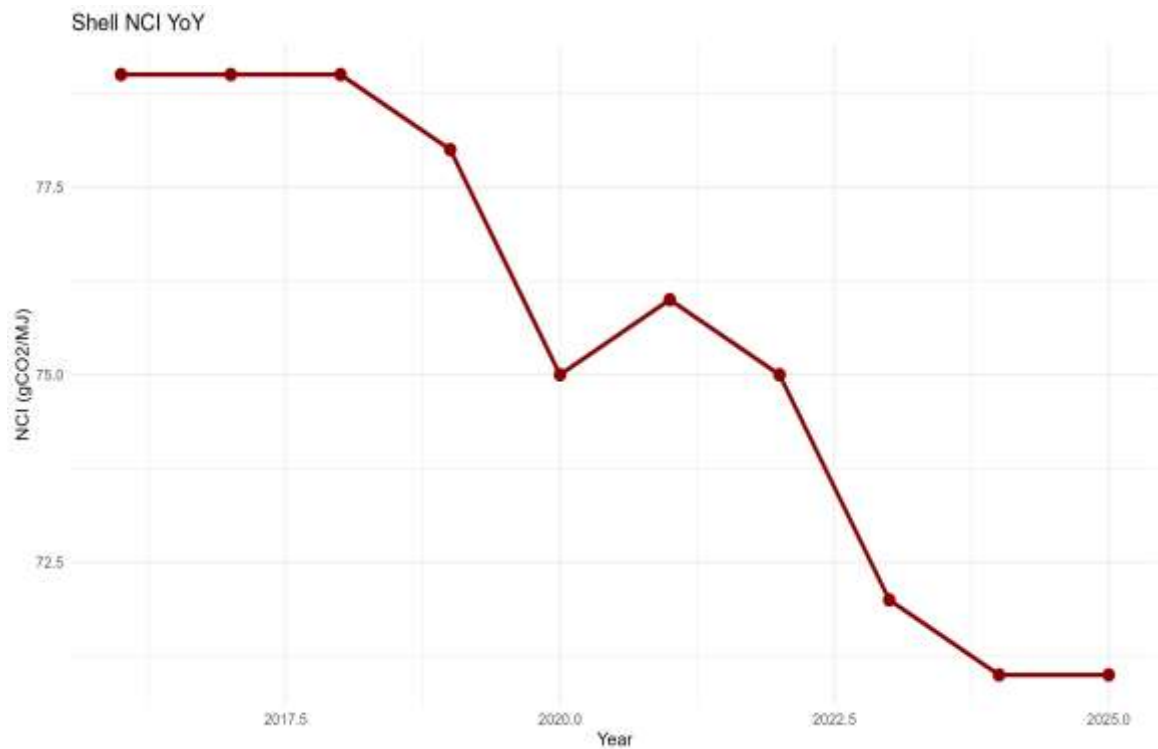


Figure 3. Net Carbon Intensity (NCI) of Sold Products (gCO₂ e/MJ), 2016–2025

Note: NCI declined 9% by 2025, on track for Shell's 15–20% by 2030 target. Improvement reflects product-mix shift toward LNG and lower-carbon fuels rather than absolute output reduction.

E2 — Scope 2 Emissions

Market-Based vs Location-Based



Figure 4. Scope 2: Market-Based vs Location-Based Emissions (MtCO₂ e), 2016–2025

Note: Market-based emissions fall faster than location-based, indicating reliance on renewable energy certificates (RECs) and power purchase agreements (PPAs) as the primary reduction mechanism.

Market-Based vs Market-Based Equity Boundary



Figure 5. Scope 2: Market-Based vs Market-Based Equity Boundary (MtCO₂ e), 2016–2025

Note: Equity Boundary Scope 2 consistently exceeds the operated boundary figure, with convergence post-2022 suggesting divestment of higher-emitting assets.

E3 — Scope 3 Emissions

Customer Use vs Products Sold (Category 11)

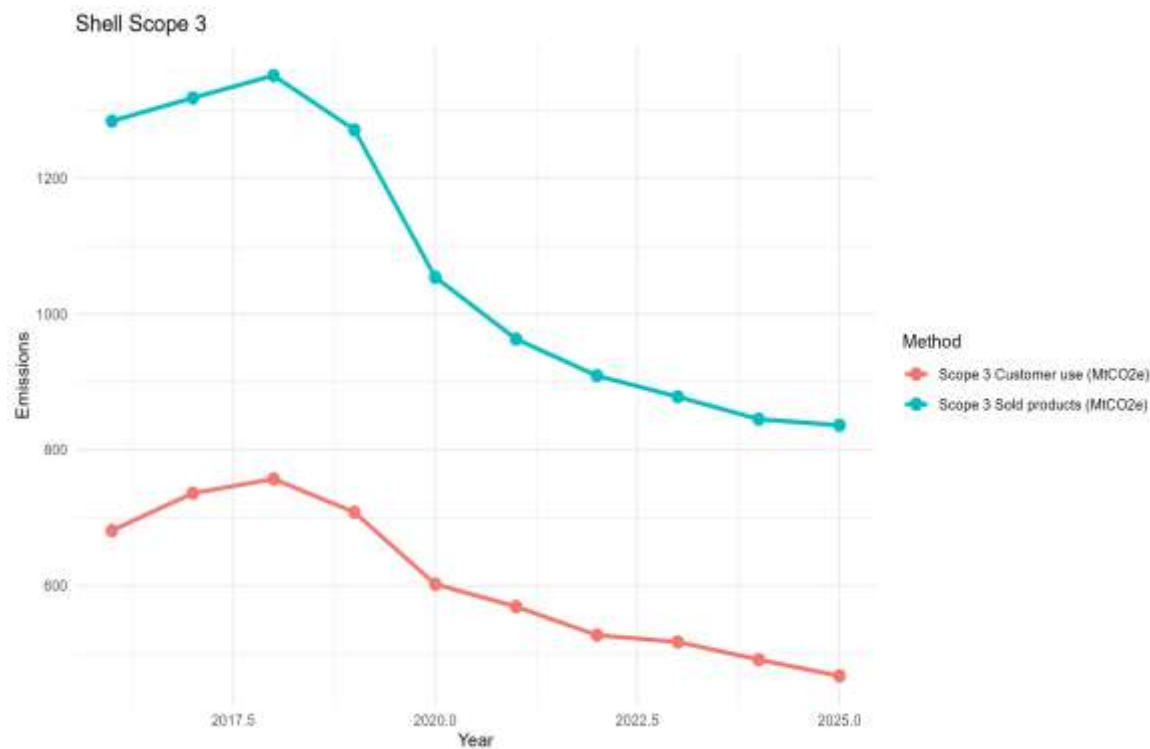


Figure 6. Scope 3 Category 11: Customer Use vs Products Sold (MtCO₂ e), 2016–2025

Note: Combined Scope 3 emissions remain the dominant component of Shell's carbon footprint. Absolute decline from the 2018 peak is real but insufficient relative to 1.5°C-aligned pathways.

E4 — Operational Environmental Indicators

Oil Spill Count

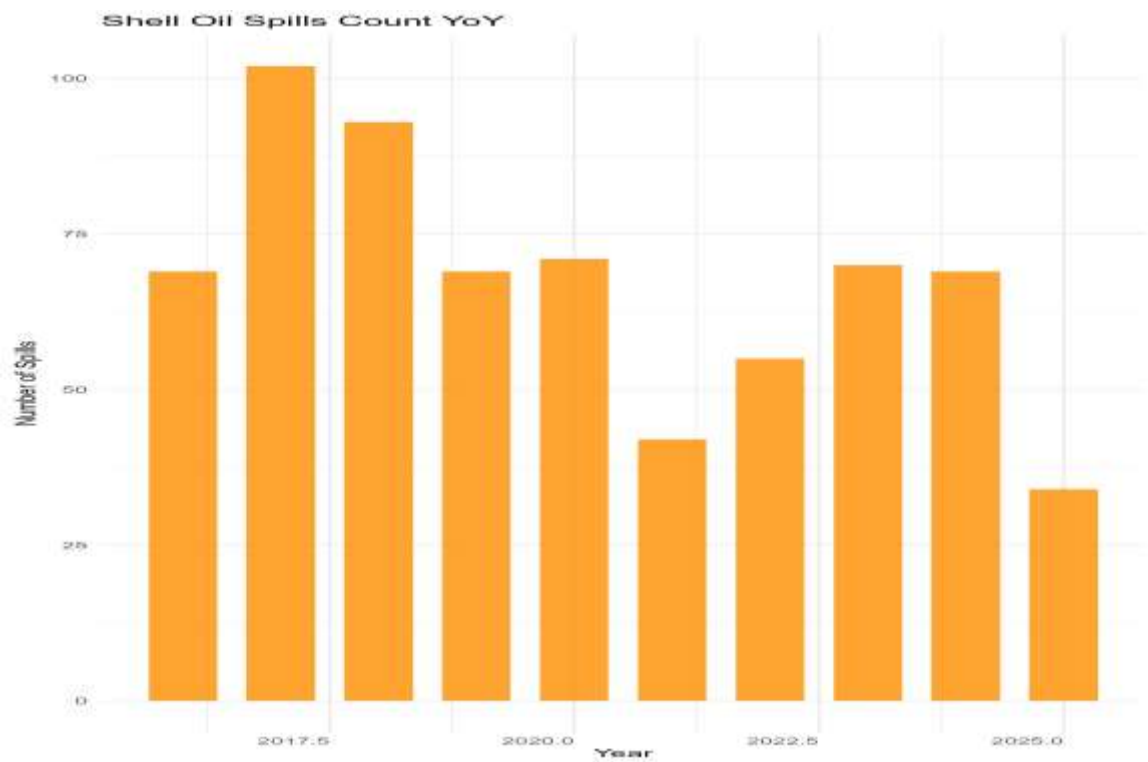


Figure 7. Oil Spill Count YoY, 2016–2025

Note: Spill frequency declined from a 2017 peak of ~105 to 33 in 2025. The plateau and rebound in 2019–2024 limits the strength of this trend.

Oil Spill Volume

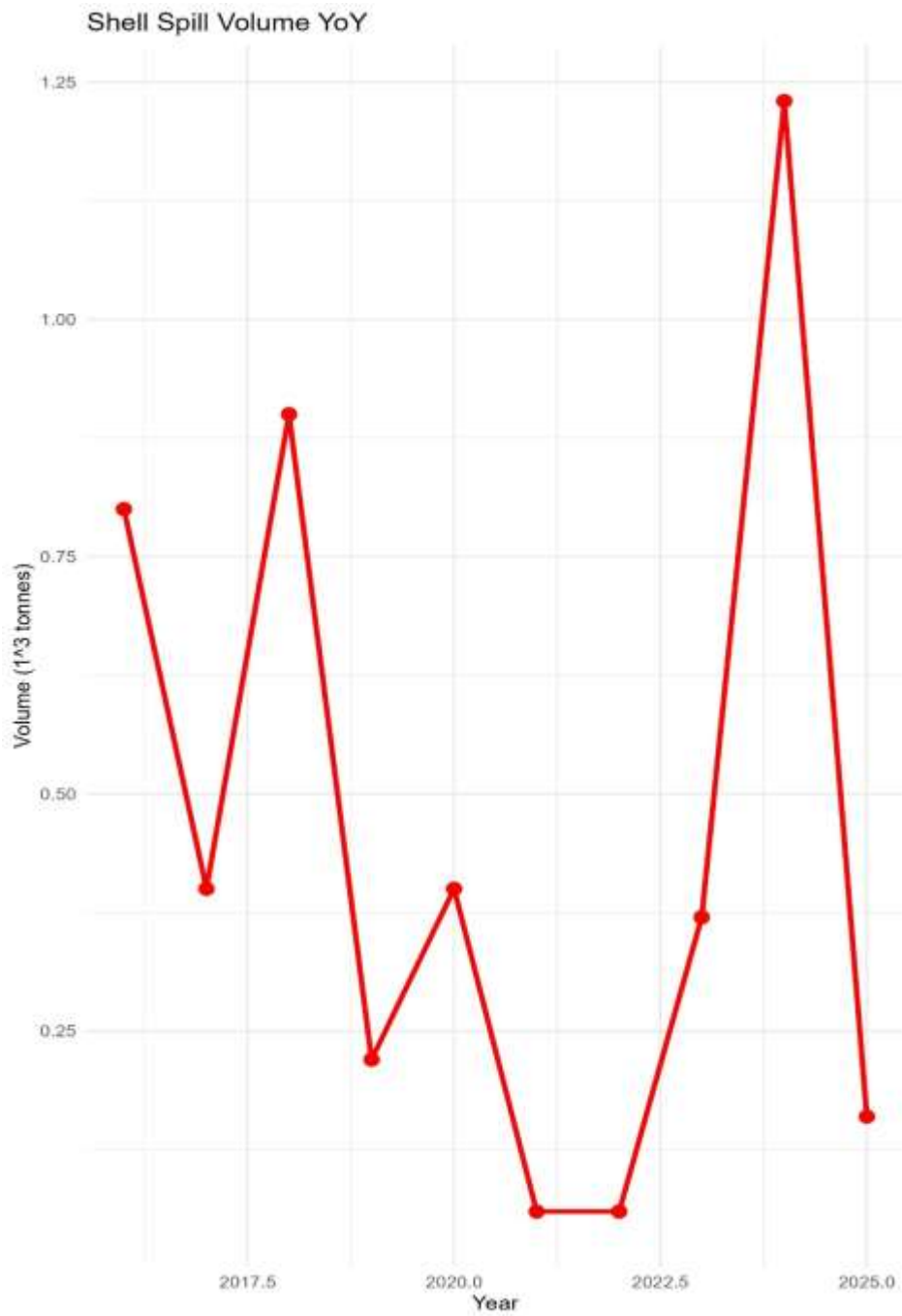


Figure 8. Oil Spill Volume YoY (1,000 Tonnes), 2016–2025

Note: Spill volume is highly volatile and does not track count. The 2024 spike to ~1,200 tonnes is the highest in the dataset — frequency metrics alone are insufficient proxies for environmental impact.

Waste: Disposed, Hazardous, Recycled

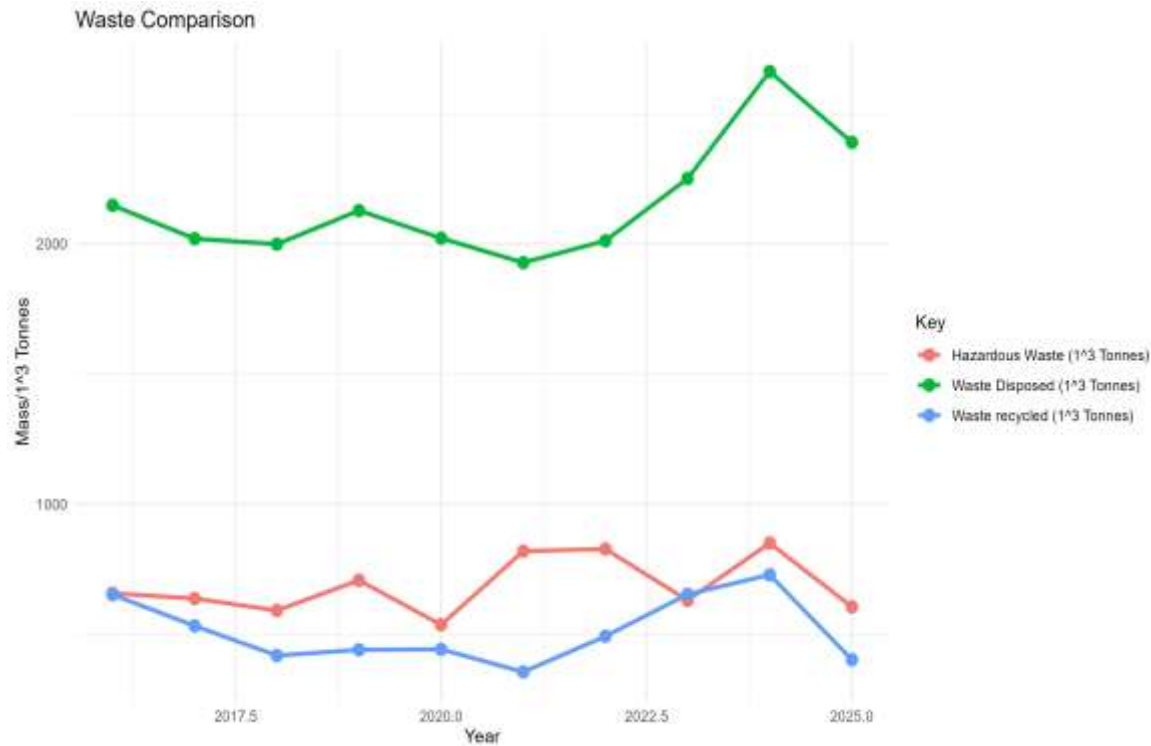


Figure 9. Waste: Disposed, Hazardous, Recycled (1,000 Tonnes), 2016–2025

Note: Waste disposed rose sharply from 2021, reaching a near-decade high by 2024. Recycling remains flat to declining. This trend contradicts broader environmental stewardship claims.

Freshwater: Withdrawn, Used, Returned

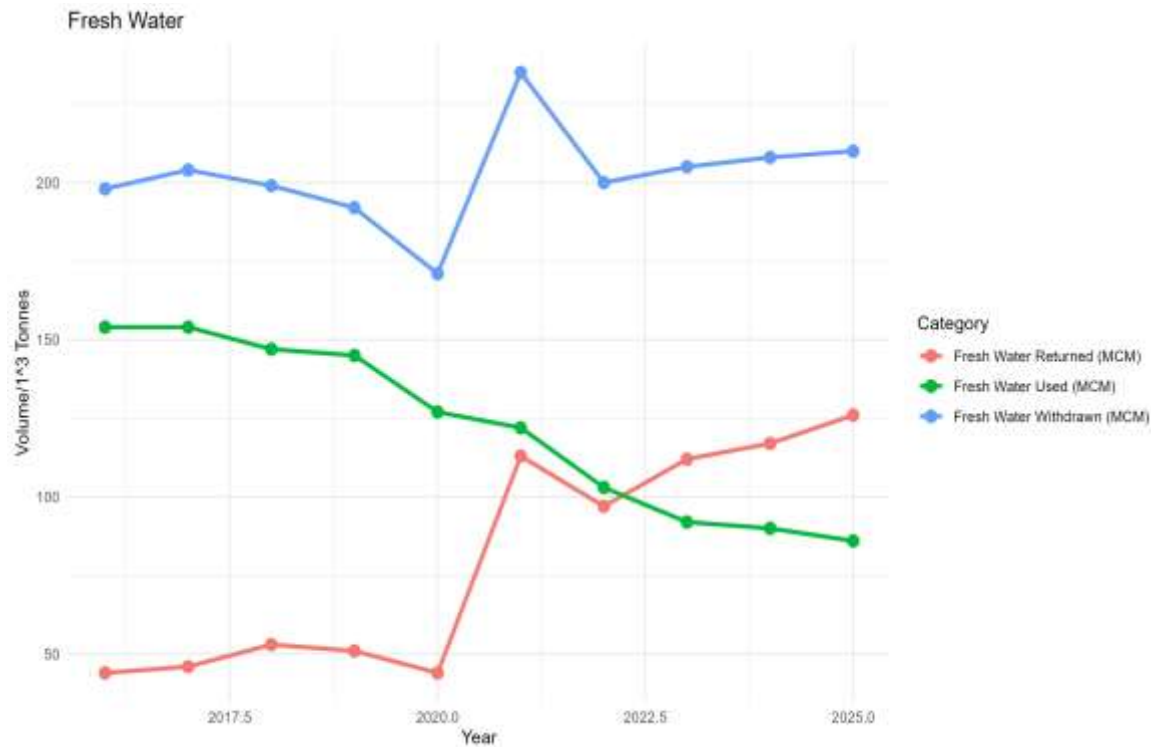


Figure 10. Freshwater: Withdrawn, Used, Returned (MCM), 2016–2025

Note: Freshwater consumed declined from ~155 MCM to ~85 MCM — the most consistently positive operational environmental indicator outside emissions. The 2021 crossover may reflect a methodology or scope change.

S1 — Social Indicators

Total Recordable Case Frequency (TRCF)

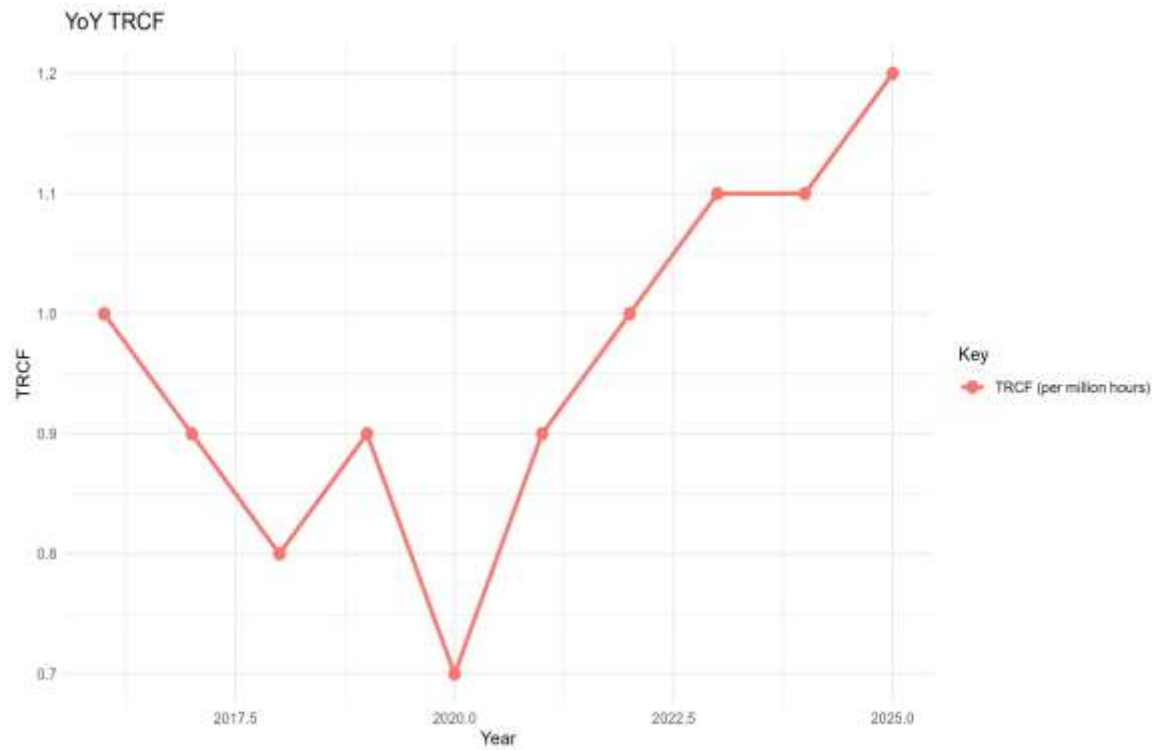


Figure 11. Total Recordable Case Frequency (TRCF) YoY, per million hours (2016–2025)

Note: TRCF reached its lowest point in 2020, almost certainly reflecting reduced operational activity during COVID-19. The consistent rise to 1.2 per million hours by 2025 — the highest in the dataset — represents a deteriorating safety frequency trend.

TRCF vs Fatalities

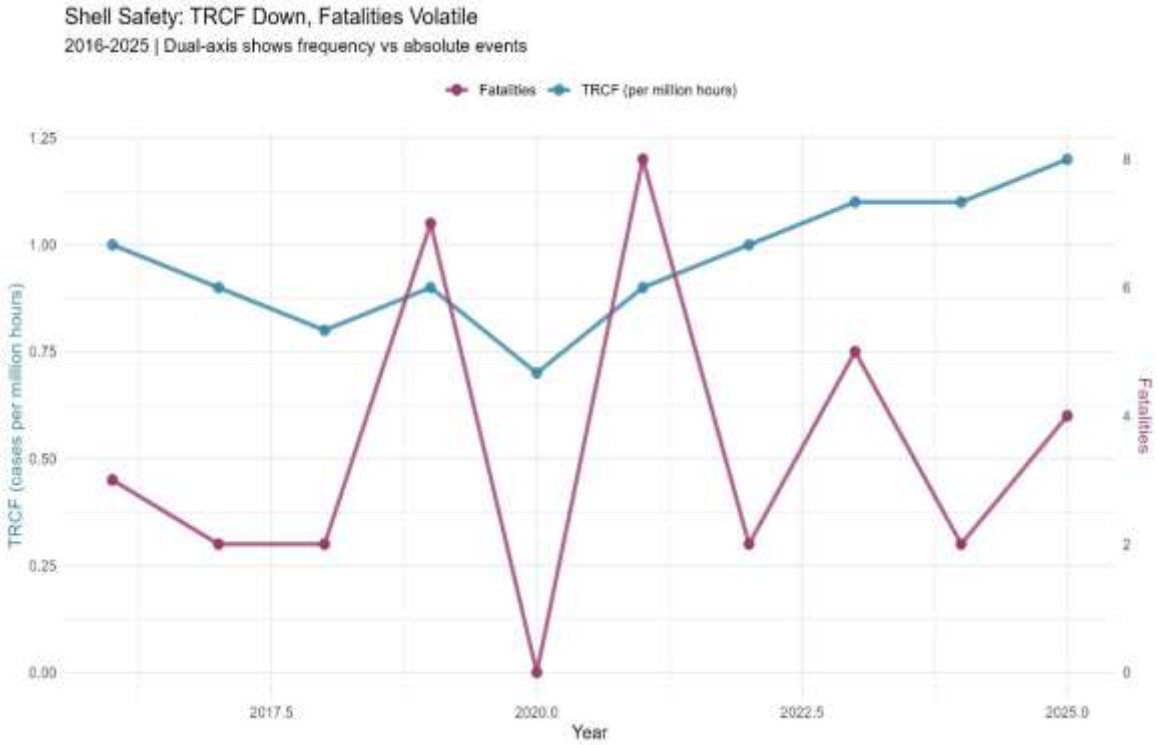


Figure 12. TRCF vs Fatalities (dual axis), 2016–2025

Note: Rising TRCF alongside volatile fatality counts post-2022 indicates the post-pandemic operational environment has not delivered safety improvement. Fatalities in 2025 (~4) are elevated relative to the pre-pandemic baseline.

Female Board Representation vs Female Workforce

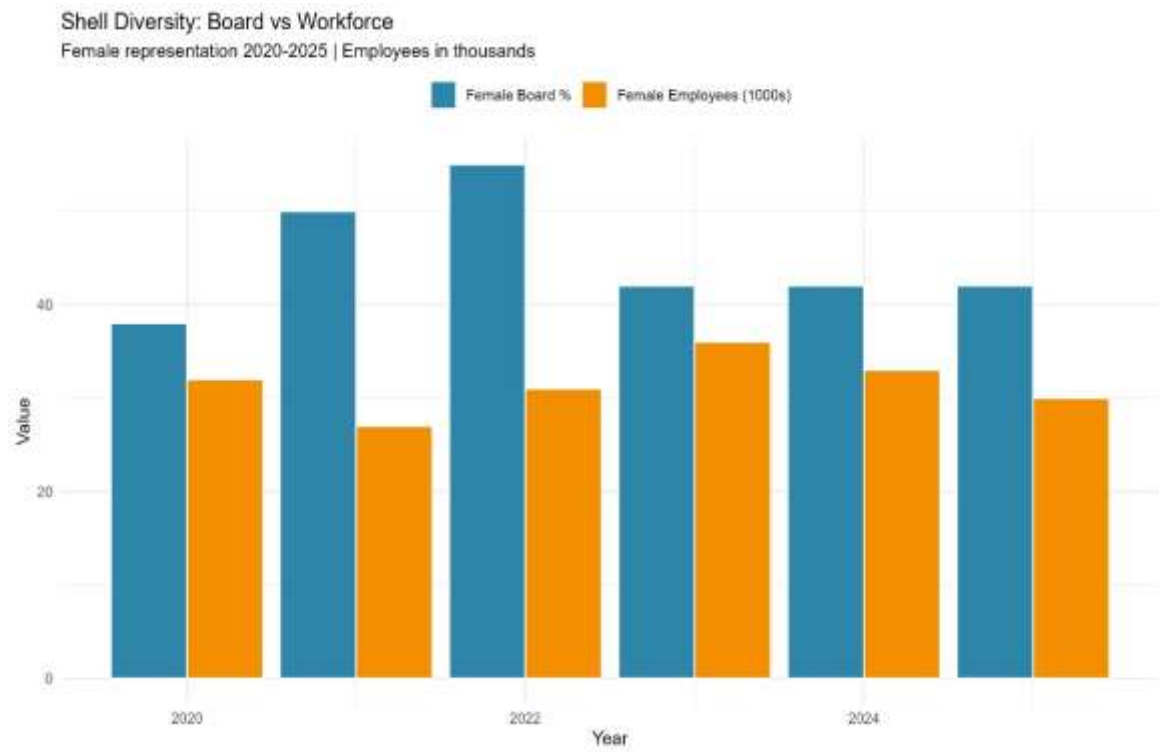


Figure 13. Female Board Representation (%) vs Female Employees (000s), 2020–2025

Note: Board-level female representation (~42%) has stabilised but stalled. Female employee headcount has declined in absolute terms since 2022. Pre-2020 data not reported by Shell.

G1 — Governance Indicators

Low-Carbon Capital Expenditure

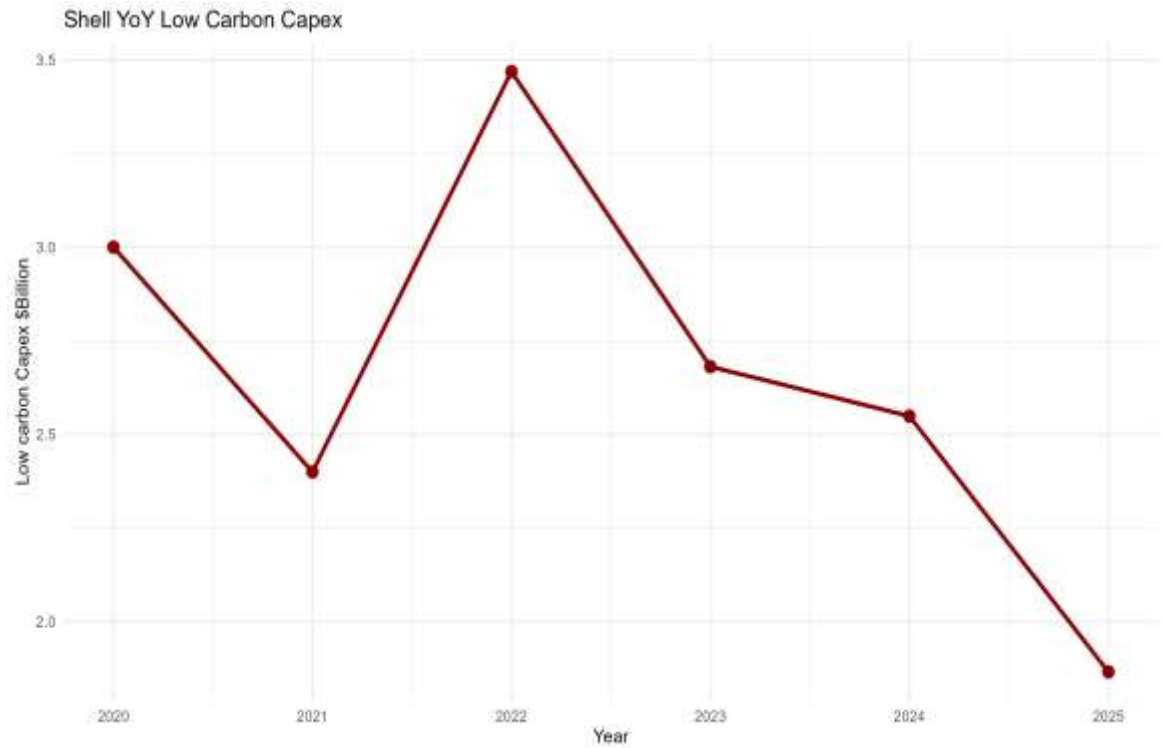


Figure 14. Low-Carbon Capital Expenditure (\$bn), 2020–2025

Note: Low-carbon capex peaked at ~\$3.45bn in 2022 and has since declined 46% to ~\$1.85bn in 2025. Mandatory UK reporting from 2020 explains the absence of pre-2020 data.

CEO-to-Median Employee Pay Ratio

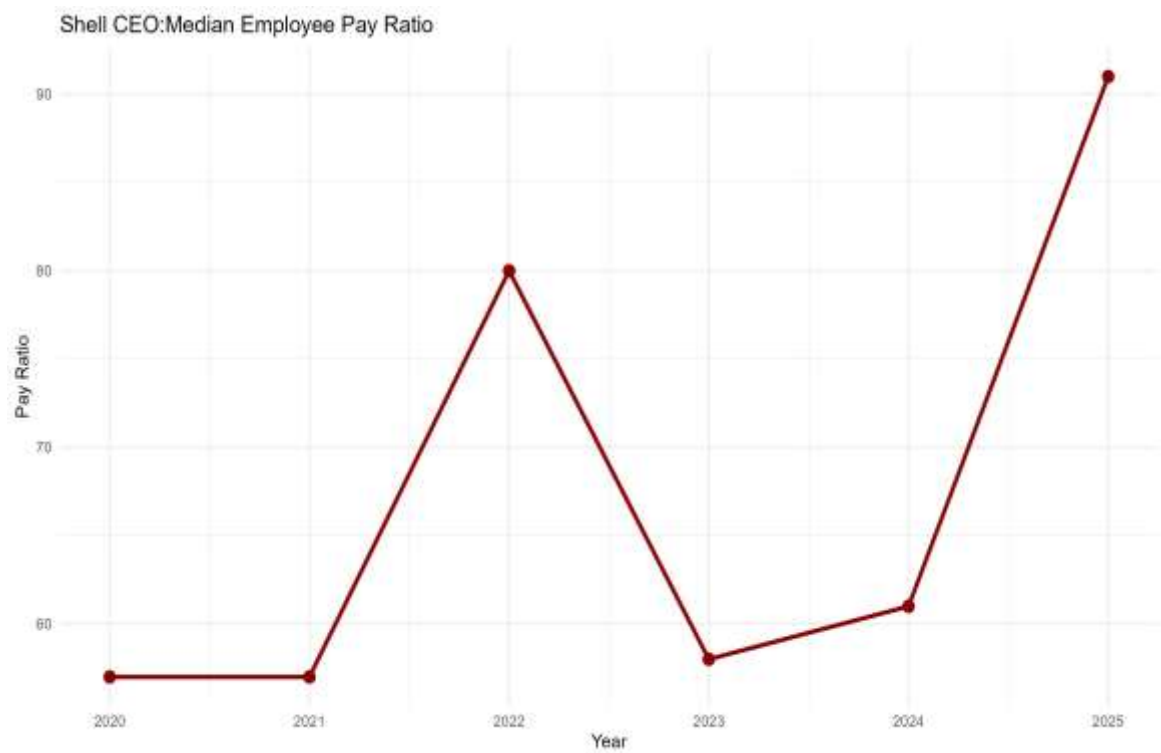


Figure 15. CEO-to-Median Employee Pay Ratio, 2020–2025

Note: The ratio reached 91x in 2025 — the highest in the dataset. Volatility reflects commodity-cycle executive bonus structures. Pre-2020 data excluded due to inconsistent voluntary disclosure prior to UK Companies Act 2019 mandatory requirements.